

Simulation Plan: Rate Allocation vs. Fixed-Rate Encoding in Decentralized Detection

Paper: Sanjari, Saldi, Gezici, Yüksel — *Decentralized Detection with Many Sensors: Optimality of Exchangeable and Identical Encoding Policies* (ICASSP 2026 short version; arXiv:2509.21724 full version)

Purpose: Meeting prep for Prof. Asaf Cohen. The paper optimizes probability of error at a *fixed* action alphabet $|\mathcal{U}|$; it contains **no communication-rate term** in J^N or J_{EE} . We add a rate-budget dimension the paper does not model, and show — using the paper's own cost metric J_{EE} and its own MAP fusion rule — that a censoring / dynamic-rate scheme (silent when uninformative) achieves a strictly better error exponent *per unit transmission budget* R . This is a **generalization along a new axis**, not a correction: the paper is not suboptimal, it simply answers a different question (error at fixed alphabet, not error per unit rate).

Positioning note (important for the oral). Scheme B's censoring policy is itself a $\star$ threshold $\succ$ policy with $|\mathcal{U}| = 3^\star$ in the sense of Definition 1: the LR values order as $\succ l(L) = 1/9 < l(O) = 1 < l(R) = 9$, and the map "0 if L , silent if O , 1 if R " assigns actions $\succ$ monotonically across the three LR cells. Hence Scheme B lives **inside** the paper's Theorem 1 (existence/optimality of threshold sensors), and Lemma 2's MAP rule remains optimal verbatim ("silent" is just another finite symbol). The **entire** novelty is the external rate accounting that makes the middle action free — nothing about the detection-theoretic structure departs from the paper.

1. Notation cross-reference (short vs. full version)

Equation numbers differ between versions; the short (ICASSP) paper omits proofs and some equations entirely.

Quantity	Short (ICASSP)	Full (arXiv)	Meaning
Expected cost $J^N(\gamma^{0:N})$	eq. (1)	eq. (2)	probability of error, N sensors
Error exponent cost, finite N : $J_{EE}^N = \log(J^N)/N$	eq. (2)	eq. (3)	primary simulated quantity
Error exponent cost, infinite N : J_{EE}	eq. (3)	eq. (4)	$\limsup_N J_{EE}^N$
MAP fusion rule / Δ_N	Lemma 2, eq. (9)-(10)	Lemma 2, eq. (11)-(12)	decoding rule implemented exactly
Threshold policy	Definition 1	Definition 1	encoding structure of both schemes
Symmetric/independent optimality (infinite pop.)	Theorem 3(i)	Theorem 5	benchmark our exponent is checked against
Representative single-sensor problem (closed form)	— (proof omitted)	eq. (26)-(28)	analytical benchmark line for Scheme A
Finite- $N \approx \infty$ - N approximation	Theorem 3(ii)	Proposition 1	convergence behavior we empirically test

Note the short version's **Theorem 3 collapses two arXiv results**: Theorem 3(i) $\approx$ arXiv Theorem 5 (existence of a symmetric/independent optimal policy) together with arXiv Proposition 1(i), and Theorem 3(ii) $\approx$ arXiv Proposition 1(ii) (ϵ_N -optimality). Because the short version omits proofs, the closed-form representative-sensor expression (eq. 26-28) only exists explicitly in the **arXiv version** — this is the anchor for the analytical benchmark line.

2. The schemes, cast in the paper's own objects

We run **three** series so the gain can be causally decomposed (see §2.1).

Scheme A — the paper's model (binary, uniform cost)

Each sensor uses a deterministic threshold policy $\gamma^i \in \Gamma_{TS}^N$ (Definition 1) with fixed $|U| = 2$ (1 bit, always transmitted, regardless of informativeness). With three observation symbols and two actions, the threshold policy must merge the uninformative O with one of $\{L, R\}$; by the $H_1 \leftrightarrow H_2$ symmetry both merges give the same cost.

Scheme A' — ternary policy, paper's uniform cost (ablation control)

The *same* ternary censoring policy as Scheme B (see below), but costed the paper's way: every symbol in $U = \{0, \text{mid}, 1\}$ costs $\log_2 3$ bits and is always transmitted, so $r_{A'} = \log_2 3$ per sensor. Same detection performance as Scheme B per sensor; **only the rate model differs**. Its role is purely to separate the two effects bundled in a naive A-vs-B contrast.

Scheme B — censoring / dynamic-rate extension

Action alphabet $U = \{\text{silent}, 0, 1\}$ with **zero rate cost assigned to silence**. A sensor transmits only when its observation is informative (L or R) and stays silent on the uninformative O . This is exactly the **censoring sensor** scheme of Rago-Willett-Bar-Shalom (1996) — see §5(c) — ported into this paper's exchangeable / error-exponent framework.

All three schemes are decoded by the **same fusion rule family** (eq. 9/11, the MAP rule proven optimal in Lemma 2, which holds for any finite U) — only the sensors' encoding policy class and the rate accounting differ, keeping the comparison apples-to-apples on the paper's own decision-theoretic terms.

2.1 Causal decomposition of the gain

- $A \rightarrow A'$ isolates the effect of the **larger alphabet** ($2 \rightarrow 3$ actions), at honest uniform cost.
- $A' \rightarrow B$ isolates the effect of the **free-silence assumption** alone (same policy, same detection curve, different rate model).

This ablation pre-empts the obvious examiner question: a bare A-vs-B contrast changes *both* alphabet size and cost model at once and cannot attribute the gain to either.

Observation model (toy, for concreteness)

Three-symbol alphabet $Y = \{L, O, R\}$, parameter $\delta \in (0, 1)$ controlling sparsity of informative observations, $a = 0.9$ fixed:

| Symbol | $P(y | H_1)$ | $P(y | H_2)$ | $l = P(\cdot | H_2) / P(\cdot | H_1)$ | --- | --- | --- | --- | L | δa | $\delta(1 - a)$ | $(1 - a)/a = 1/9$ | O | $1 - \delta$ | $1 - \delta$ | 1 | R | $\delta(1 - a)$ | δa | $a/(1 - a) = 9$ |

O is uninformative ($l = 1$); L, R are informative and symmetric under $H_1 \leftrightarrow H_2$.

3. Independent variable: total transmission budget R , not N

What R measures. R is a **transmission / channel-use / energy budget** — expected number of transmitted bits summed over sensors — **not** a mutual-information budget. This distinction is the whole ballgame:

- "Silence is free" is a statement about the **channel/energy** cost: a silent sensor consumes no slot and no energy. It is **not** a claim that silence carries no information — in an activity-detection model the fusion center still learns that a silent sensor's LR fell

in the censored region. The scheme exploits cheap *channel usage*, not zero information.

- Consequently the per-sensor rate is the **expected transmitted bits**:

$$r_A = 1, \quad r_{A'} = \log_2 3, \quad r_B(\delta) = \delta.$$

- **Caveat to flag to Asaf:** this accounting assumes activity detection is itself free (no reservation minislot per sensor). If detecting "sensor i is active" costs a slot, r_B picks up an $O(1)$ per-sensor term and the $1/\delta$ advantage shrinks. The unsourced / activity-detection channel model is exactly the regime where $r_B = \delta$ is the right bookkeeping.

****Why not fix N :** comparing at fixed N silently hands Scheme B a smaller total budget for free and fails to isolate *efficient use of a fixed budget*. Fix R , derive the number of sensors each scheme can afford:

$$N_A(R) = \frac{R}{r_A} = R, \quad N_{A'}(R) = \frac{R}{\log_2 3}, \quad N_B(R) = \frac{R}{r_B(\delta)} = \frac{R}{\delta}.$$

Then run each scheme's own simulation at *its own* $N(R)$ and compare $\hat{J}_{EE}^{N(R)}$ at matched R . This is the fair axis for the headline plot.

Where the gain actually comes from (narrative for the writeup). At matched R , Schemes A and B make the *same* expected number of transmissions: Scheme A has R sensors all transmitting (R transmissions); Scheme B has R/δ sensors each transmitting w.p. δ (also $\approx R$ transmissions). So the gain is **not "more bits"** — it is that Scheme B's transmissions are *selected* to be the informative ones (L/R), while Scheme A spends its bits on average observations that are usually the uninformative O . In one line: **the budget buys informative transmissions instead of average ones**. This is the censoring insight, and Rago et al. prove the optimal censoring region is a single LR interval — precisely the middle cell we silence.

4. Estimating J^N : exact (counts / convolution) first, sampling only as a cross-check

J^N (eq. 1/2) is a probability of error:

$$J^N = P(H_1)P(u^0 = 2 \mid H_1) + P(H_2)P(u^0 = 1 \mid H_2).$$

Key simplification (do this first). Because sensors are i.i.d. under each hypothesis and the policy is symmetric, Δ_N (eq. 10/12) depends **only on the counts** of each action value, and each per-sensor log-LR term takes finitely many values. Two exact routes, both far cheaper and more accurate than Monte Carlo:

- **Exact via action-count vectors:** enumerate count vectors $(n_1, \dots, n_{|U|})$ with $\sum n_k = N - O(N^{|U|-1})$ of them (i.e. $O(N^2)$ for the ternary schemes) — weight each by its multinomial probability under each H_j , apply the exact MAP rule, sum. Exact, no sampling noise, polynomial in N .
- **Exact via LLR convolution (preferred for large N):** the per-sensor log-LR $\log[P(u|H_1)/P(u|H_2)]$ is a finite discrete random variable, so the exact law of $\sum_i \text{LLR}_i$ (hence Δ_N , hence J^N) is the N -fold convolution of that law — computable by FFT, essentially linear in N , exact to arbitrarily large N .

This **removes the rare-event problem** that would otherwise force a large-deviations tier: we get the exact tail directly. It also **fixes an enumeration-ceiling error**: naive *sequence* enumeration is $|U|^N$, i.e. 3^N for the ternary schemes (feasible only to $N \approx 15$, not the 20-25 that holds for the binary 2^N case). The counts/convolution method sidesteps this entirely.

- **Monte Carlo (independent cross-check only):** draw $y^{\wedge\{1:N\}}$ i.i.d. under H_j , form $u^i = \gamma^i(y^i)$ and Δ_N , apply the MAP threshold, record errors over $T \sim 10^5$ - 10^6 trials; standard error $\approx \sqrt{\hat{J}(1 - \hat{J})/T}$. Use it to validate the exact code in a regime where errors are still observable, **not** as the production estimator (plain MC needs $T \gg 1/J^N$ and dies in the large- N tail).

Plan: exact counts/convolution as the production estimator across all N ; Monte Carlo agreement in the moderate- N overlap as a sanity check on the implementation.

5. Benchmarks (reference lines)

Series on one R -axis plot:

(a) **Scheme A exact points** — should converge, as $N_A(R) \rightarrow \infty$, to (b).

(b) **Paper's closed-form asymptotic exponent for the representative sensor** (arXiv eq. 26).

The benchmark **line is an error exponent**, so it is the Chernoff-information form and the priors $P(H_j)$ do **not** appear in it:

$$J_{EE}^A = \min_{s \in [0,1]} \log \sum_{u \in U} g(H_1, u; \gamma^R)^s g(H_2, u; \gamma^R)^{1-s},$$

minimized over the (few) representative threshold policies γ^R , with $g(H, u; \gamma^R) = \sum_y \gamma^R(u | y) P(y | H)$.

Correction vs. previous draft. The earlier benchmark formula applied an outer log to the $>$ arXiv eq. (28) expectation-of-exp form *and* retained the $P(H_j)$ priors — that is neither eq. $>$ (26) nor eq. (28) cleanly. Eq. (28), $\min_s \sum_j E_{P_\pi^R}[\exp(s \log \frac{g(H_1, u)}{g(H_2, u)}) | H_j] P(H_j)$, $>$ is a **finite- N error bound**, in which the priors are sub-exponential prefactors; its $>$ normalized log-limit $\frac{1}{N} \log(\cdot)$ reproduces the exponent in eq. (26), with

the priors > washing out. For a benchmark *line*, use eq. (26) as above. This is the primary sanity check: if the Scheme A exact points don't converge to this line, the MAP implementation has a bug.

(c) **Tsitsiklis's original large- N bound** [21 short / 27 arXiv] — the classical identical-LRT-sensor result the paper builds on and cites in the Introduction and in its own Appendix 8.7 proof. Weaker than (b) but establishes lineage — useful for the "relationship to prior work" section.

(c') **Censoring lineage (prior work, for the writeup — not a plotted exponent).** The send/no-send mechanism is **not novel**; it is the censoring sensor scheme:

- C. Rago, P. Willett, Y. Bar-Shalom, "Censoring sensors: a low-communication-rate scheme for distributed detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 32, no. 2, pp. 554–568, 1996. **Main result:** under a rate constraint, to minimize error probability a sensor should transmit **iff its local LR falls outside a single interval** — i.e. the optimal censoring region is one contiguous LR interval, exactly our "silent on O ." So Scheme B's structure is provably optimal, but the mechanism is classical.
- S. Appadwedula, V. V. Veeravalli, D. L. Jones, "Decentralized detection with censoring sensors," *IEEE Trans. Signal Process.*, vol. 56, no. 4, pp. 1362–1373, 2008; and "Energy-efficient detection in sensor networks," *IEEE J. Sel. Areas Commun.*, vol. 23, no. 4, pp. 693–702, 2005.

Ahlsvede-Csiszár (ref. [18 short]/[24 arXiv]) is the right lineage for "detection under a rate constraint" in general; censoring is the specific mechanism. Honest positioning: **Scheme B ports classical censoring into the paper's exchangeable / error-exponent setting**, not a new detection idea.

(d) **Scheme A' and Scheme B exact points + Scheme B's analytical exponent line.** The A' series isolates the alphabet effect; the B series adds free silence. Scheme B's asymptote is $-C_{\text{raw}}(\delta)$ in matched sign convention (see §5.1) — **our proposed extension's line**, not from the paper.

5.1 Sign convention (fix once, use everywhere)

J_{EE} is a log-probability of error ($\rightarrow -\infty$ as $N \rightarrow \infty$), while Chernoff information C is a positive decay rate. Under matched definitions $J_{EE} = -C$, consistent across all series.

5.2 Analytical gain (closed form — use as the reference curve in fig3)

Small- δ asymptotics for the toy model give per-sensor Chernoff information

$$\text{Gain}(\delta) \approx \frac{1 - 2\sqrt{a(1-a)}}{\max_s h(s)} \cdot \frac{1}{\delta}$$

At matched R : Scheme A total exponent $\approx R, C_A = O(R\delta)$; Scheme B total exponent $\approx$

$(R/\delta)\backslash, C_B = R\backslash \big(1-2\sqrt{a(1-a)}\big) = O(R)$. Hence which is $\Theta(1/\delta)$ with an **a -dependent constant**. At $a = 0.9$: $\max_s h(s) \approx 0.20$ (near $s \approx 0.6$) and $1 - 2\sqrt{a(1-a)} = 0.4$, giving $\text{Gain} \approx 2/\delta$. Report the result as $\text{Gain}(\delta) = \Theta(1/\delta)$ with the constant shown — " $2/\delta$ " is the $a = 0.9$ special case, not universal. (fig3) should trace this closed form, not just " $\approx 2/\delta$."

6. Deliverables

****Axis correction (do not plot normalized J_{EE} on the R -axis).**** The headline gain lives in the *sensor count* ($N_B = R/\delta \gg N_A = R$), and the paper's normalized exponent $J_{EE}^N = \frac{1}{N} \log J^N \rightarrow -C^{\text{ps}}$ **divides that count back out**: it converges to a $>$ per-sensor constant that is independent of both R and the rate r . Plotting $J_{EE}^{N(R)}$ vs. $> R$ would therefore show only the $\times 2$ per-sensor informativeness gain and **hide the $> 1/\delta$ count gain entirely**. The headline metric must be the **unnormalized total exponent** $E(R) := -\log J^N(R) \approx \frac{C^{\text{ps}}}{\delta} R$ whose slope in R is C^{ps}/r . The gain is the **slope ratio** $> \text{slope}_B / \text{slope}_A = C_B^{\text{ps}} / (\delta C_A^{\text{ps}}) \approx 2/\delta$. (Normalized J_{EE} is still the correct y -axis for fig2, which is vs. N for a single scheme.)

- (sim_JEE.py) — implements Sections 2-5 and 8; exact log-domain estimator, MC cross-check, all three scheme series.
 - (fig1_totalexponent_vs_R.png) — **main plot**: total exponent $E(R) = -\log J^{N(R)}$ vs. R for Schemes A, A', B (three lines; slope ratio B/A is the gain), with the closed-form slopes from §5.2 and §8.7 overlaid. Optionally add a twin axis showing the actual error probability $J^{N(R)}$.
 - (fig2_JEE_vs_N.png) — convergence check: J_{EE}^N (normalized, this axis is correct here) vs. N for Scheme A only, confirming convergence to benchmark (b), validating Theorem 3(ii)/Proposition 1 empirically.
 - (fig3_gain_vs_delta.png) — gain (slope ratio) of Scheme B over Scheme A at matched R vs. δ , overlaid on the closed-form reference of §5.2 (not just a heuristic $2/\delta$ line).
 - Headline summary table (printed): per-scheme rate r , per-sensor Chernoff info C^{ps} , slope C^{ps}/r , and the $A \rightarrow A'$ vs. $A' \rightarrow B$ decomposition of the gain, at a chosen operating point (e.g. $\delta = 0.1, R = 20$ bits).
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7. Open items to confirm before/at tomorrow's meeting

- Sign convention $J_{EE} = -C$ fixed (§5.1) — confirm applied uniformly before plotting.
- Confirm exact counts/convolution estimator agrees with Monte Carlo in the moderate- N overlap (implementation sanity check).
- Frame Scheme B to Asaf as a **generalization along a new (rate) axis**, sitting inside the paper's Theorem 1 as a $|U| = 3$ threshold policy — **not** a correction of the paper.

- Credit the **censoring lineage** (Rago-Willett-Bar-Shalom 1996; Appadwedula-Veeravalli-Jones 2005/2008) in the "relationship to prior work" section; the mechanism is classical, our contribution is the error-exponent / exchangeable-framework port.
 - Flag explicitly: the Scheme B gain rests on **silence being free at the channel/energy level** (unsourced/activity-detection model). If activity detection costs a per-sensor slot, the $1/\delta$ advantage shrinks.
 - Decide whether to also add a Gaussian-shift variant for a more "continuous observation" feel (the discrete three-symbol model already suffices for the headline).
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8. Implementation specification (build target for Claude Code)

Single file `sim_JEE.py`, dependencies `numpy`, `scipy` (`scipy.special.gammaln`, `logsumexp`, `scipy.optimize.minimize_scalar`), `matplotlib`. **Everything error-probability-related is computed in the log domain** (see §8.9) — never materialize J^N as a float, it underflows around $NC \gtrsim 700$.

8.1 Config

```
a          = 0.9
priors     = (0.5, 0.5)          # (P(H1), P(H2))
deltas    = [0.02, 0.05, 0.1, 0.2, 0.4]
R_grid    = np.linspace(2, 40, 20) # transmission-budget axis (bits)
N_max_exact = 4000              # O(N^2) log-domain cap; beyond this, use closed-form
line only
MC_trials  = 200_000           # cross-check only
```

8.2 Per-scheme action distributions (exact; functions of `a`, `delta`)

Return, for each hypothesis, a dict `{action: prob}`.

- **Scheme A** (binary, policy $L \rightarrow 0, \{O, R\} \rightarrow 1$):
 - $H_1: P(0) = \delta a, P(1) = 1 - \delta a$
 - $H_2: P(0) = \delta(1 - a), P(1) = 1 - \delta(1 - a)$
- **Ternary policy** (shared by A' and B; $L \rightarrow 0, O \rightarrow \text{mid}, R \rightarrow 1$):
 - $H_1: P(0) = \delta a, P(\text{mid}) = 1 - \delta, P(1) = \delta(1 - a)$
 - $H_2: P(0) = \delta(1 - a), P(\text{mid}) = 1 - \delta, P(1) = \delta a$

A' and B have **identical** $J^N(N)$ — they differ only in rate r (hence in $N(R)$). Compute the ternary error curve once and reuse.

8.3 Per-sensor log-LR values (nats)

$v(u) = \log[P(u \mid H_1)/P(u \mid H_2)]$.

- Scheme A: $v(0) = \log \frac{a}{1-a}$, $v(1) = \log \frac{1-\delta a}{1-\delta(1-a)}$ (two distinct reals).
- Ternary: $v(0) = \log \frac{a}{1-a}$, $v(\text{mid}) = 0$, $v(1) = -\log \frac{a}{1-a}$ (lattice $\pm\beta$, 0 with $\beta = \log \frac{a}{1-a}$).

8.4 Exact error probability in log-domain $\rightarrow$ returns $\log J^N$

MAP threshold on the summed log-LR $S = \sum_i v(u^i)$: **decide** H_1 **iff** $S \geq t_S$, where $t_S = \log \frac{P(H_2)}{P(H_1)}$ (ties $\rightarrow H_1$; $t_S = 0$ for equal priors). Then

$$J^N = P(H_1) P(S < t_S | H_1) + P(H_2) P(S \geq t_S | H_2).$$

- **Scheme A (binomial, exact, $O(N)$):** $n_0 \sim \text{Binomial}(N, P(0 | H_j))$, $n_1 = N - n_0$, $S(n_0) = n_0 v(0) + n_1 v(1)$. For each hypothesis, `logsumexp` the log-binomial terms $[\log \binom{N}{n_0} + n_0 \log p_0 + n_1 \log p_1]$ over the error region for that hypothesis. Use `gamma1n` for $\log \binom{N}{n_0}$.
- **Ternary A'/B (multinomial counts, exact, $O(N^2)$):** iterate (n_L, n_R) with $n_{\text{mid}} = N - n_L - n_R \geq 0$; $S = (n_L - n_R)\beta$; `logsumexp` the log-multinomial terms $[\log \frac{N!}{n_L! n_{\text{mid}}! n_R!} + n_L \log p_L + n_{\text{mid}} \log p_{\text{mid}} + n_R \log p_R]$ over the error region. (Optimization: S depends only on $D = n_L - n_R$, so accumulate log-mass by D ; still exact.)

Signature: `log_JN(scheme, N, a, delta, priors) -> float # returns  $\log J^N$  (natural log)`.

8.5 Normalized and total exponents

- `J_EE(scheme, N, ...) = log_JN(...) / N` (per-sensor; $\rightarrow$ benchmark (b) as $N \rightarrow \infty$).
- `total_exponent(scheme, N, ...) = -log_JN(...)` (**headline metric**, unnormalized).

8.6 Budget-axis mapping

`r = {'A': 1.0, 'Aprime': np.log2(3), 'B': delta}`;
`N_of_R(scheme, R, delta) = max(1, round(R/r))`. For each R in `R_grid`: if $N(R) \leq \text{N_max_exact}$ use exact `total_exponent`; else use the closed-form slope line (§8.7). Plot $E(R)$ vs R .

8.7 Benchmarks (closed forms)

Chernoff info of a representative sensor with action law $g(H, u)$:

$$C^{\text{ps}} = - \min_{s \in [0,1]} \log \sum_u g(H_1, u)^s g(H_2, u)^{1-s} \quad (\text{nats}),$$

computed with `minimize_scalar` on $[0, 1]$ (log-domain summand via `logsumexp` of $s \log g_1 + (1-s) \log g_2$).

- **(b) Scheme A asymptote:** $C_A^{\text{ps}}(\delta)$ from the binary $g = P(u | H)$. `(fig2)` points must converge to $-C_A^{\text{ps}}$.
- **(d) Scheme B asymptote:** $C_B^{\text{ps}}(\delta)$ from the ternary g (the mid/silent term contributes $(1 - \delta)$ to the Chernoff sum).
- **Slopes for fig1:** $\text{slope}_{\text{scheme}} = C^{\text{ps}}/r$.
- **(c) Tsitsiklis line:** for this binary toy it **coincides with (b)** (identical 1-bit LRT sensor); include it only as an annotation for the "prior work" narrative, or omit from the plot to avoid a duplicate line. Do **not** present it as an independent tighter/looser curve here.
- **Gain (fig3):** $\text{Gain}(\delta) = \frac{C_B^{\text{ps}}(\delta)}{\delta C_A^{\text{ps}}(\delta)}$ (exact), overlaid with the small- δ closed form $\frac{1 - 2\sqrt{a(1-a)}}{\max_s h(s)} \cdot \frac{1}{\delta}, h(s) = [sa + (1-s)(1-a)] - a^s(1-a)^{1-s}$.

8.8 Monte Carlo cross-check (validation only)

`mc_JN(scheme, N, a, delta, priors, trials)` — sample $y^{\{1:N\}}$ under each H_j , map to actions, compute S , apply the MAP threshold, return empirical $\hat{J}^N$ with its standard error. Run only at moderate N where $J^N \gtrsim 10^{-4}$; assert agreement with `(exp(log_JN(...)))` within ~ 3 standard errors.

8.9 Numerical-stability requirements (mandatory)

- All probability sums via `(scipy.special.logsumexp)`; all factorials via `(gammaLn)`. Never call `(math.comb)`/`(factorial)` on large N , never exponentiate before summing.
- `(log_JN)` returns natural log; convert to log2/bits only at display time if desired.
- Chernoff minimization on a bounded scalar $s \in [0, 1]$; guard $g = 0$ entries (drop them, since $0^s = 0$) before taking logs.

8.10 Acceptance tests (assert in a `(__main__)` self-check)

1. **MAP sign:** with equal priors, threshold $t_S = 0$; flipping $H_1 \leftrightarrow H_2$ leaves J^N invariant (model is symmetric) — assert `(log_JN)` unchanged under the swap.
2. **Convergence:** `(J_EE('A', N, ...))` $\rightarrow$ `(-C_A_ps)` within tol as N grows (drives fig2).
3. **Exact vs. MC:** §8.8 agreement at moderate N .
4. **Binary-policy symmetry:** the alternative Scheme A merge ($\{L, O\}$ vs $\{R\}$) yields the same C_A^{ps} — assert equal, confirming the merge choice is immaterial.
5. **A' $\equiv$ B detection:** `(log_JN('Aprime', N, ...) == log_JN('B', N, ...))` for all N (they share the ternary curve) — the only difference must appear via `(r)/(N_of_R)`.
6. **Gain scaling:** fitted slope of $\log \text{Gain}(\delta)$ vs. $\log \delta \approx -1$ (confirms $\Theta(1/\delta)$), and the constant matches §5.2 at $a = 0.9$ (≈ 2) within tol.

8.11 Public function summary

```
action_dists(scheme, a, delta) -> (dict_H1, dict_H2)
llr_values(scheme, a, delta)  -> dict{action: v(u)}
log_JN(scheme, N, a, delta, priors) -> float          # exact, log-domain
J_EE(scheme, N, a, delta, priors)  -> float          # = log_JN/N (normalized)
total_exponent(scheme, N, a, delta, priors) -> float  # = -log_JN (headline)
chernoff_ps(scheme, a, delta)  -> float              # benchmark (b)/(d)
gain(delta, a)                  -> float              # slope ratio B/A
mc_JN(scheme, N, a, delta, priors, trials) -> (Jhat, se)
```

8.12 Scope reminder for the implementer

The middle/silent action is uninformative by construction ($v = 0$): it must contribute $(1 - \delta)$ to Chernoff sums and 0 to S . The **only** thing that distinguishes A' from B anywhere in the code is the per-sensor rate $\boxed{r}$; if any detection-side quantity differs between them, that is a bug.